FC^2P: Feature Cross-Channel Projection for Unsupervised Anomaly Segmentation

Yichi Chen, Weizhi Xian, Junjie Wang, *Graduate Student Member, IEEE*, Xian Tao, *Senior Member, IEEE*, and Bin Chen

***Abstract*—Unsupervised anomaly segmentation plays a critical role in real-world industrial product quality inspection. While feature reconstruction-based methods have shown promising performance by detecting anomalies through differences between pretrained features and their reconstructions, existing approaches often suffer from shortcut learning, and leading to reconstruction failures and inaccurate anomaly representation across multistage features. To address these limitations, we propose feature cross-channel projection (FC^2P), a novel approach for anomaly segmentation. FC^2P divides features into two subsets based on neighboring channels and employs two autoencoders for closed-loop prediction, effectively mitigating shortcut effects while capturing semantic relationships for efficient reconstruction. In addition, we introduce an anomaly exposure network (AExNet), which progressively amplifies anomalies across multistage feature residuals, generating precise anomaly score maps for accurate segmentation. Extensive experiments on MVTec AD and Visa benchmark datasets demonstrate that the proposed FC^2P achieves state-of-the-art (SOTA) performance, with average precision (AP) scores of 79.8% and 44.8%, respectively. Moreover, visualization results on real industrial data further show the practicality of our proposed method. The code will be made publicly available at https://github.com/Karma1628/work-2 to ensure reproducibility and facilitate further research.**

***Index Terms*—Anomaly detection, anomaly segmentation, feature cross-channel projection (FC^2P), feature reconstruction, self-supervised learning.**

I. Introduction

ANOMALY segmentation aims at detecting pixels that deviate from normal behavior within an image. Many challenging and complex visual tasks, including medical diagnosis [1], [2], industrial defect detection [3], and autonomous driving [4], necessitate anomaly detection. In practical applications, abnormal samples are extremely scarce, while the abnormal categories are often unpredictable. To this end, many researchers from both industry and academia [5], [6], [7] conduct anomaly segmentation in an unsupervised setting, where only normal samples are available for training. The mainstream approaches to tackle this problem encompass embedding-based methods [8], [9] [10], knowledge distillation-based methods [11], [12] [13], and reconstruction-based methods [14], [15] [16], [17].

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The proposed approach in this study falls under the category of reconstruction-based methods, which dominate unsupervised anomaly segmentation. Reconstruction-based methods implicitly learn the latent distribution of normal data, using autoencoders [17], [18], generative adversarial networks [14], [19], and denoising diffusion probabilistic models [20], [21], through the minimization of reconstruction loss during training. At inference time, these methods assume that the latent manifold learned exclusively from normal samples cannot effectively reconstruct abnormal regions. Therefore, the reconstruction residual, calculated as pixelwise differences between the reconstructed result and the original input, serves to represent the pixel anomaly scores.

A promising subbranch of reconstruction-based methods involves reconstructing multistage features [7], [23], [24] extracted from normal images using a pretrained network. Previous reconstruction-based methods primarily operate in image space, where slight reconstructive perturbations of pixels produce reconstruction errors comparable to those induced by real anomalies, failing to meet the basic assumption. As shown in the first row and third column of Fig. 1, for the image reconstruction method DRAEM [6], the failure to accurately reconstruct the texture details of the carpet produces numerous false positives (red circles) near actual cutting anomalies. In contrast, pretrained features are inherently more informative and discriminative, which increases the representational distance between normal and abnormal regions, thereby mitigating the aforementioned interference. As illustrated in the first row and fourth column of Fig. 1, feature reconstruction method DFR [7] produces almost no false positives near the anomaly in the carpet.

However, feature reconstruction-based methods still face two primary challenges in practical applications. One is common in reconstruction schemes, where the reconstructive

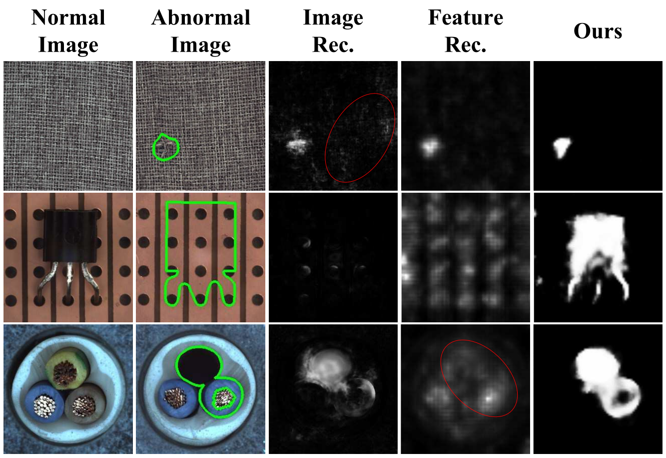

Fig. 1. Qualitative results of image reconstruction-based method DRAEM [6], feature reconstruction-based method DFR [7] and the proposed FC^{2}P on the MVTec AD [22] dataset for anomaly segmentation. The categories are carpet, transistor, and cable (top to bottom). The interior regions of green boundaries in the second column denote abnormal areas from actual binary segmentation masks. Compared with previous methods, our proposed method not only significantly enhances the saliency of anomaly segmentation but also effectively suppresses false alarms in normal backgrounds.

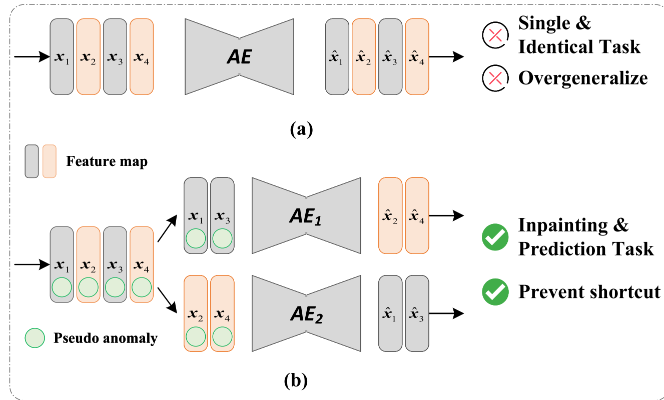

Fig. 2. Different feature reconstruction frameworks. (a) Conventional identical feature reconstruction, where the reconstructive target treats the input data and the target as identical, allowing the autoencoder to take shortcuts and merely replicate the input. (b) Our proposed FC^{2}P uses the similarity of neighborhood channel feature maps, enabling the autoencoder to perform two nonidentical proxy tasks: inpainting the pseudo-anomaly and predicting the neighborhood channel feature maps, thereby mitigating the aforementioned issue.

network sometimes *overgeneralizes*, resulting in anomalies in features also being well reconstructed. This occurs because the reconstruction network's output is identical to the input during training, leading to shortcuts where the network merely replicates the original input. The second row of Fig. 1 illustrates the issue of false negatives caused by overgeneralizing, as previous reconstruction-based methods, both in image and feature space, reconstruct the background instead of the missing transistor. Another challenge is that anomalies can be *overwhelmed* in multistage features, thus preventing effective segmentation and leading to missed detections. Each feature stage contains information at different scales and semantic levels, leading to varying emphasis on different anomalies. Intuitively, summing the feature reconstruction residuals over channels to represent the anomaly score for each spatial location is inadequate. In DFR [7], four stages of features suffice for detecting pixel-level anomalies in the texture-based tile, but as deeper features are introduced, the performance deteriorates. In addition, this issue also manifests in situations where anomalies are suppressed by other anomalies. As shown in the third row and fourth column of Fig. 1, the missing anomaly of the cable is severely suppressed by the color change anomaly (red circle) in DFR [7]. This is because the latter anomaly accounts for a low proportion in the feature reconstruction residuals, which cannot be revealed in the anomaly score map by traditional summation methods.

To address the aforementioned challenges, we propose feature cross-channel projection along with an anomaly exposure network (AExNet), called FC^{2}P, a novel feature reconstruction framework for anomaly segmentation and detection. It is observed that the feature map of each stage is derived from the abstraction and nonlinear combination of all feature maps from the previous stage. Therefore, the intuition behind our method is that there is a potential mapping relationship between the feature maps of the neighborhood channels, governed by the structure of the pretrained network. This motivates the use of deep autoencoders to learn this mapping, as an additional normal attribute to fully leverage the structural characteristics of the pretrained network. Fig. 2(a) and (b) illustrates the framework differences between conventional identical feature reconstruction and our proposed FC^{2}P, respectively. Previous methods aggregate multichannel feature maps into a unified representation, focusing on scale differences [25], [26] or improving network architecture [27], [28] [29] to enhance reconstruction quality. However, they often suffer from shortcut issues due to the reconstruction target being identical to the input. In contrast, our method uses deep autoencoders to simultaneously perform two nonidentical prediction tasks, effectively mitigating this problem. Specifically, normal images are superimposed with synthetic patterns to simulate pseudo-anomalies, so the extracted features are actually corroded according to certain rules. The features are then split into two nonoverlapping feature subsets based on neighborhood channels. Due to the intrinsic correlation between neighborhood channel feature maps, autoencoders can be employed for closed-loop prediction from one feature subset to another to ensure the integrity of the features. In addition, the autoencoder must not only predict neighborhood channel relationships but also complete the pseudo-anomaly inpainting task. Inpainting anomalies in features helps to reduce the impact of noise and redundant information, while also allowing anomalies to vary in scale and shape, thus enhancing the robustness of reconstruction.

Furthermore, considering that anomalies can be overwhelmed in the multistage representation space of features, we design an AExNet to adaptively estimate the anomaly score map from neighborhood feature reconstruction residuals. The network is trained under the popular self-supervised framework [6], [30], guided by pseudo-abnormal binary masks, and has been shown to generalize well for detecting real anomalies. Specifically, the key component is the Siamese anomaly propagator (SAP), which leverages the similarity between neighborhood subsets to obtain enhanced common representations of anomalies and propagates them across stages. Notably, the feature reconstruction residuals are divided into aggregated and multiscale morphologies, which are decoded using the U-Net structure. This design, inspired by the divide-and-conquer approach, not only balances the representation of anomalies across the entire feature space but also gradually enhances anomalies across hierarchical stages.

Extensive experiments conducted on various benchmark datasets demonstrate the effectiveness of our approach. Notably, the proposed method achieves the state-of-the-art (SOTA) anomaly segmentation performance with an average precision (AP) metric of 79.8% on the widely used MVTec AD [22] dataset and 44.8% on the challenging Visa [31] dataset. We also conduct comprehensive ablation studies and visualizations to validate the effectiveness of our proposed components. The main contributions of our proposed method are summarized as follows.

1) We propose FC^{2}P, a new paradigm for feature reconstruction. The inherent relationship among the neighborhood feature maps enables the autoencoder to accomplish two nonidentical prediction tasks, thereby mitigating the issue of over generalization in the conventional reconstruction scheme.
2) We propose an AExNet to adaptively estimate an anomaly score map based on the similarity between neighborhood reconstruction residuals. Its design inherits the divide-and-conquer principle, which is conducive to simultaneously perceiving anomalies in features both globally and hierarchically.
3) Extensive experiments on several datasets demonstrate the superiority of our proposed FC^{2}P over several SOTA methods for both unsupervised anomaly segmentation and detection tasks. The visualization results in real-world scenarios also demonstrate the practicality of our method.

II. Related Works

A. Image Reconstruction-Based Methods

Reconstruction-based methods assume that the reconstructive network trained on normal samples generates higher reconstruction errors on abnormal regions. Therefore, anomaly segmentation can be achieved through the discrepancy between the test image and its reconstruction. AEs [15], [17] and GANs [14], [19], [32], both of which follow the encoder–decoder framework, are commonly employed reconstructive networks. Moreover, leveraging the inherent prior information in the image can enhance reconstruction performance. For example, EdgeRec [33] reconstructs the original RGB image from its grayscale edge, incorporating skip connections in the autoencoder to preserve high-frequency information. To improve EdgeRec, ensuring consistency between the image and its structure, P-Net [17] incorporates the encoded latent space of the image into the structural latent space. Furthermore, it has been demonstrated that using GANs and DDPMs can improve generation results [34], [35]. For instance, GANomaly [32], one of the first GANs used for anomaly detection, learns representations in both image and latent vector space jointly in an adversarial training process. To achieve superior mode coverage compared with GANs, AnoDDPM [21] proposes a multiscale simplex noise diffusion process that controls the target size of pseudo-noises to detect larger anomalies.

However, deep autoencoders tend to "generalize" so well that they directly replicate abnormal areas in their reconstruction, resulting in misdetections of anomalies. To address this issue, MemAE [15] introduces an updatable memory module with an attention mechanism to enhance the latent space, encouraging reconstruction to approach the normal prototype in memory. MemSTC-Net [16] combines the advantages of [15] and [17], dynamically storing structure–texture information in the memory bank. Nevertheless, MemSTC-Net performs poorly in reconstructing details because the memory module constrains the representation capacity of the autoencoder. A viable solution is to reconstruct an image, whose partial regions are randomly removed guided by masks, in an inpainting manner as in [36]. To generate diverse masks, SCDAN [37] introduces multiple scale and strip directions in them to encourage the reconstructive network to learn the semantic context from different locations, scales, and directions. However, during inference, SCDAN requires iterating over all combinations of masks. To improve inference efficiency, SSM [5] designs a progressive mask refinement approach, which iteratively refines masks and focuses them on the abnormal regions based on reconstruction errors.

Although image reconstruction-based methods are straightforward, the limited and uniform nature of normal samples restricts the representations that autoencoders can learn, making them inadequate for handling complex anomalies. This article introduces inpainting pseudo-anomalies within discriminative pretrained features, thereby compelling the autoencoder to learn diverse normal representations.

B. Feature Reconstruction-Based Methods

Deep feature extractors [38], [39] trained on large-scale datasets [40] have become integral components in anomaly segmentation methods. Generalized pretrained features, derived from diverse natural images [41], generate more discriminative representations for normal images, thereby increasing the discrepancies between normal and abnormal data. Embedding-based methods [8], [9], [42] and knowledge distillation-based methods [11], [13], [43] are among the pioneering approaches in applying pretrained features in anomaly detection tasks. The embedding-based methods aim to model the pretrained region descriptors in statistical approaches, such as multivariate Gaussian distributions [9] and normalizing flows [10], to explicitly quantify deviations of anomalies within the distribution. Knowledge distillation-based methods [11], [13], [44] assume that the student network cannot imitate the behavior of anomalies in the pretrained teacher network. However, these methods are highly sensitive to the configuration of pretrained networks, including the selection of stages, network structure, and distribution design [41].

In contrast, feature reconstruction-based methods are more effective at capturing the latent space distribution of normal pretrained features and offer greater flexibility in structural

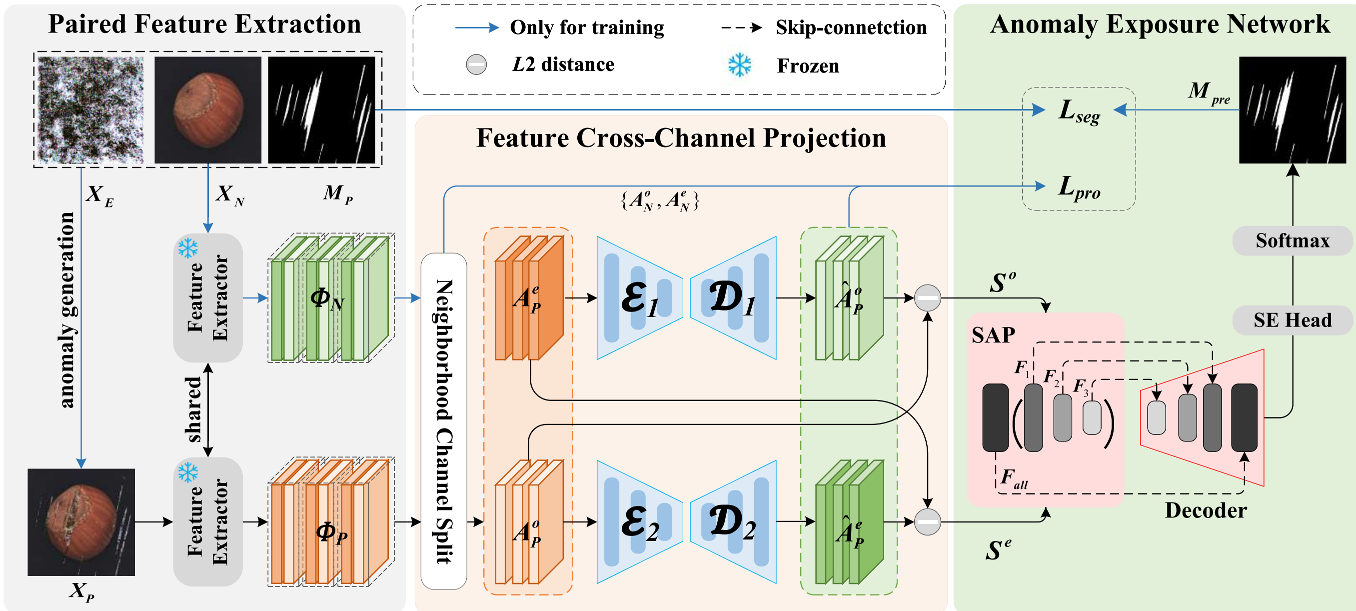

Fig. 3. Overview of FC^{2}P. During the training phase, pseudo-abnormal images and their corresponding normal images are fed into the pretrained feature extractor to obtain multistage aggregated features. Two autoencoders separately project subsets of features from the pseudo-abnormal feature, which are divided by neighborhood channel split, onto normal feature subsets, thereby accomplishing proxy tasks for neighborhood prediction and inpainting. The feature reconstruction residual subsets are then fed into the AExNet that is trained under the supervision of pseudo-abnormal masks. During the testing phase, the data flow of test images is kept consistent with that of the pseudo-abnormal images.

design. DFR [7] first proposed using convolutional autoencoders to reconstruct multiscale pretrained features with a single forward pass, achieving smooth segmentation results. ADTR [23] introduces a transformer-based autoencoder to address the identity mapping issue in convolutional networks. UTRAD [24] integrates skip connections and pyramid structures in the transformer-based autoencoder to detect multiscale structural and nonstructural anomalies.

However, because the reconstruction target is identical to the original input data, autoencoders tend to uncontrollably replicate input behavior. In contrast, our proposed approach enables the reconstruction network to effectively circumvent the identity mapping process by accomplishing two proxy tasks: inpainting pseudo-anomalies in features and projecting across feature channels, thereby enhancing its reconstructive capabilities for structure and semantic information. The latter primarily relies on the fact that the cross-channel feature maps are combined by the unique structure of the pretrained network, thus motivating us to exploit the potential correlation between them.

C. Self-Supervised Learning Approaches

Recent self-supervised learning methods have yielded promising results in anomaly segmentation by introducing pseudo-abnormal samples to learn the boundaries between abnormal and normal pixels. CutPaste [45] generates pseudo-anomalies by cutting and pasting parts of normal images and then employs a one-class classifier to distinguish them. However, for anomaly segmentation, the use of Grad-CAM [46] in CutPaste can only roughly locate anomalies. Subsequently, DRAEM [6] proposes leveraging a discriminative network with U-Net [47] to learn pixel differences between the pseudo-abnormal image and its inpainted in a self-supervised manner. This not only reduces overfitting to pseudo-anomalies but also enhances the ability to segment abnormal regions with finer granularity. To reduce dependency on external data during the synthesis of pseudo-anomalies in [6], DSR [48] samples discrete latent space features to generate pseudo-anomalies, improving the detection of anomalies near the distribution. Furthermore, DiffAD [49] employs latent diffusion models [50] to improve image reconstruction performance and proposes interpolated channels to increase reconstruction diversity. The discriminative network can also be integrated with other methods, extending beyond image reconstruction, to segment discrepancies. For instance, MemSeg [30] introduces a memory module that stores multistage normal features to guide the decoding process of the discriminative network through an attention mechanism. DeSTSeg [12] proposes a segmentation network that perceives feature differences between the denoising student network and the fixed teacher network. The denoising student network is trained on pseudo-abnormal samples to generate feature representations similar to those of the teacher network. To further improve the student network's feature representation of abnormal regions, DAF [51] introduces auxiliary loss heads to supervise each stage of the student network.

However, these works focus on processing overall differences and do not fully consider the propagation of anomalies across different feature stages. Although DAF uses auxiliary losses to supervise the stages of the student network, it still simply aggregates their results, making the anomaly score map coarse. In contrast, our proposed method adopts the idea of divide and conquer, not only considering the overall performance of anomalies in aggregated features, but also allowing anomalies to progressively enhance across stages.

III. Methods

This section elaborates on the principle of our proposed FC^{2}P, whose training framework is illustrated in Fig. 3. The model consists of three cascaded components: paired feature extraction, FC^{2}P, and an AExNet. Specifically, during the training phase, pseudo-abnormal images and their corresponding normal images are fed into a weights-shared pretrained network to extract aggregated multistage discriminative features. Subsequently, the features are divided into two subsets through neighborhood channel split, achieving closed-loop prediction by cross-channel feature projection. The novelty lies in projecting the pseudo-abnormal feature subset onto the neighborhood normal feature subset, thus avoiding the replication of original input behavior caused by identical reconstruction. Finally, the AExNet accurately segments anomalies, which could be overwhelmed in the feature reconstruction residuals, for effectively producing the precise anomaly score map.

A. Paired Feature Extraction

The training of our model requires pseudo-abnormal images synthesized on normal samples as self-supervised signals. In this work, the synthesis algorithm for pseudo-abnormal images is consistent with [6], which involves utilizing a binarized 2-D perlin noise map as a mask to alpha blend external texture data from [52] onto normal images. Formally, in the training phase, given a normal image $X_N \in \mathbb{R}^{H\times W\times C}$, where H, W, and C denote the dimensions of image height, width, and channels, its pseudo abnormal image X_P is generated at the same time using a binary pseudo-mask $M_P \in \mathbb{R}^{H\times W\times 1}$ and an external texture image X_E. The synthesis process can be expressed as follows:

$$X_P = X_N \odot \overline{M_P} + \alpha\,(X_N \odot M_P) + (1-\alpha)\,(X_E \odot M_P) \tag{1}$$

where $\overline{M_P} = 1 - M_P$, $\odot$ is the elementwise product, and $\alpha \in (0, 0.8]$ denotes the opacity factor, controlling the degree of pseudo-anomalies to increase diversity.

To obtain more discriminative representations, a frozen pretrained network [53] with four different stages is employed to extract multistage features from images. Considering that the features from the last stage are more abstract and highly correlated with the training task, they are typically discarded in fine-grained downstream tasks [54]. Let $\phi_i \in \mathbb{R}^{H_i\times W_i\times C_i}$ be the feature map extracted from stage i, where H_i, W_i, and C_i denote the height, width, and channels of the feature map, and $i = 1, 2, 3$. Subsequently, these feature maps from different stages are aggregated using a simple strategy, which involves resizing them to the resolution of the largest feature map (H_1, W_1) and then concatenating them along the channel dimension. Therefore, by aggregating the feature maps of the first three stages (ϕ_1, ϕ_2, ϕ_3) extracted from X_N and X_P, the normal feature $\Phi_N \in \mathbb{R}^{H_1\times W_1\times C_{\text{sum}}}$ and pseudo-abnormal feature $\Phi_P \in \mathbb{R}^{H_1\times W_1\times C_{\text{sum}}}$ can be obtained in pairs, where $C_{\text{sum}} = C_1+C_2+C_3$. Intuitively, aggregating multistage feature maps can integrate both low-level and high-level semantic representations of both normal and abnormal images. Compared with DeSTSeg [12], where features are reconstructed hierarchically, this strategy can facilitate the interaction of information across different stages.

B. Feature Cross-Channel Projection

The proposed FC^{2}P is motivated by the observation of the features extracted via the pretrained network. In pretrained networks, it is evident that the feature map $x_i^j \in \mathbb{R}^{H_i\times W_i}$ of stage i with channel index j originate from the abstraction and nonlinear combination, which is denoted as $f_j(\cdot)$, of the features ϕ_{i-1} from the preceding stage $i-1$. Then, the feature map x_i^j and its neighborhood feature map x_i^{j+1} can be formulated as follows:

$$x_i^j = f_j(\phi_{i-1}), \quad x_i^{j+1} = f_{j+1}(\phi_{i-1}) \tag{2}$$

where f is typically determined by the structure of the pretrained network. Obviously, there exists an inherent correlation between the neighborhood feature maps. The correlation $F_{\cdot\mapsto\cdot}(\cdot)$ can be presented as follows:

$$\begin{aligned} x_i^j &= F_{j+1\mapsto j}\left(x_i^{j+1}\right) = f_j^{-1}\left(f_{j+1}^{-1}\left(x_i^{j+1}\right)\right)\\ x_i^{j+1} &= F_{j\mapsto j+1}\left(x_i^{j}\right) = f_{j+1}^{-1}\left(f_j^{-1}\left(x_i^{j}\right)\right). \end{aligned} \tag{3}$$

This insight inspires us to leverage deep networks to realize the mutual projection across channels. Furthermore, under the introduction of pseudo-anomalies, this projection also encompasses an inpainting function.

Specifically, for the feature Φ, feature maps of neighborhood channels are divided into two subsets $A^e \in \mathbb{R}^{H_1\times W_1\times (C_{\text{sum}}/2)}$ and $A^o \in \mathbb{R}^{H_1\times W_1\times (C_{\text{sum}}/2)}$, referred to as neighborhood channel split. The above process can be presented in detail as follows:

$$\begin{aligned} A^e &= \text{cat}\left(\left\{\Phi^k \mid k = 2j\right\}\right)\\ A^o &= \text{cat}\left(\left\{\Phi^k \mid k = 2j-1\right\}\right) \end{aligned} \tag{4}$$

where $\Phi^k \in \mathbb{R}^{H_1\times W_1\times 1}$ is the feature map of channel index k in feature Φ and $j = (1, 2, \ldots, (C_{\text{sum}}/2))$. cat$(\cdot)$ denotes the concatenation along the channel dimension. Therefore, following the neighborhood channel split of Φ_N and Φ_P, the normal neighborhood feature subsets (A_N^e, A_N^o) and pseudo-abnormal neighborhood feature subsets (A_P^e, A_P^o) can be obtained.

To explore the latent feature distribution of normal samples, the convolutional autoencoder is used to accomplish two proxy tasks: inpainting pseudo-anomalies and predicting the neighborhood feature map. Specifically, an encoder is employed to compress features into a latent space, followed by a decoder that maps this latent space back to the corresponding normal neighborhood features. The projection process can be formulated as follows:

$$\begin{aligned} \hat{A}_N^e &= \mathcal{D}_1\left(\mathcal{E}_1\left(A_P^o\right)\right)\\ \hat{A}_N^o &= \mathcal{D}_2\left(\mathcal{E}_2\left(A_P^e\right)\right) \end{aligned} \tag{5}$$

where $\hat{A}_N^e, \hat{A}_N^o \in \mathbb{R}^{H_1\times W_1\times (C_{\text{sum}}/2)}$ are the projected features, $\mathcal{E}_1, \mathcal{E}_2$ and $\mathcal{D}_1, \mathcal{D}_2$ denote the encoders and decoders used for the projection process, respectively. Given that neighborhoods

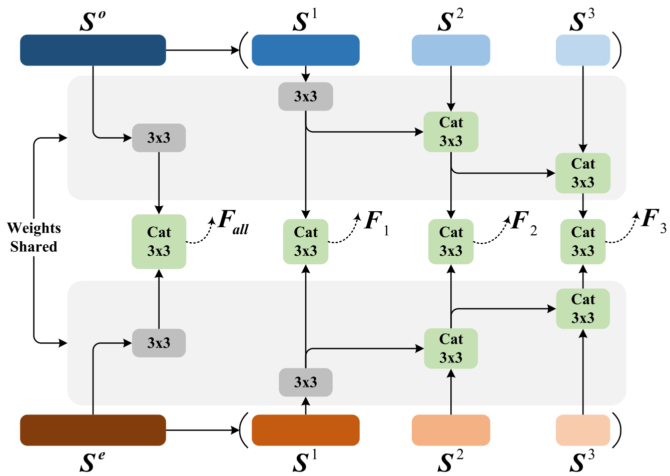

Fig. 4. SAP of AExNet for self-supervised learning. Not only the aggregated feature reconstruction residuals are fused, but the representation of anomalies in their multistage feature sets is encoded with shared weights and integrated with the previous stage by other convolutional blocks.

are mutual and nonoverlapping, two autoencoders are required to complete the closed-loop mapping, which can concurrently maintains the integrity of the projected feature.

Finally, reconstruction-based anomaly segmentation methods typically employ pixelwise reconstruction errors to describe the severity of anomalies. For the features, a simple approach is to directly use the elementwise L_2 distance between the original input and its reconstruction as follows:

$$S^e(i,j,k) = \|\hat{A}_N^e(i,j,k) - A_N^e(i,j,k)\|_2$$
$$S^o(i,j,k) = \|\hat{A}_N^o(i,j,k) - A_N^o(i,j,k)\|_2 \tag{6}$$

where $S^e, S^o \in \mathbb{R}^{H_1 \times W_1 \times (C_{\text{sum}}/2)}$ are feature reconstruction residuals, and i, j, and k denote the index of height, width, and channel, respectively. As the distance increases, it indicates that the region is more likely to be abnormal.

C. Anomaly Exposure Network

Anomalies at each spatial position can be manifested through the collective element discrepancies across channels. However, features typically vary in both resolution and number of channels for stages, making the direct summation of errors along the channel dimension suboptimal. To address this issue, we propose an AExNet, as shown in Fig. 3, inspired by the divide-and-conquer approach. Specifically, considering that anomalies behave various in different stages, resulting in being overwhelmed in aggregation, a SAP is further proposed to allow discrepancies, enhancing through the stages as illustrated in Fig. 4. The aggregated feature reconstruction residual is split into three subsets according to their original stages, and their resolutions are simultaneously restored. Hence, for S^e and S^o, their three-stage feature subsets $\{S_1^e, S_2^e, S_3^e\}$ and $\{S_1^o, S_2^o, S_3^o\}$ can be obtained, where the subscript numbers denote the stage they belong but their channel count is only half of the original. Notably, the reacquisition of multiscale information is pivotal for anomaly segmentation practices.

Since neighborhood features possess inherent correlations, they typically exhibit high similarity in response to anomalies. Therefore, this similarity can be utilized to calibrate and filter out the noise information. For S^e and S^o with their feature subsets, weights shared 3×3 convolutional blocks are utilized to learn consistent feature representations of anomalies. Furthermore, to obtain unified feature reconstruction residuals, the neighborhood features encoded with shared weights are concatenated and integrated with the previous stage by other convolutional blocks. These blocks can calibrate the unstable offsets in neighborhood features and encode these features into more compact features F_{all}, F_1, F_2, and F_3, where F_{all} denotes the fused aggregated feature and others are fused features of different stages.

Finally, the fused compact features from SAP are gradually incorporated into the decoding process through skip connections according to their resolution. Interestingly, anomalies propagate from lower to deeper stages in SAP, while the decoding process supports the propagation of anomalies from low to high resolution. This ensures that anomalies are exposed in each feature stage. Furthermore, the aggregated F_{all} are connected to the last decoder with F_1, following a squeeze-and-excitation (SE) [55] head to adaptively weight the feature maps for predicting the precise anomaly score mask M_{pre}.

D. Self-Supervised Loss Functions

For the FC^{2}P, the training objective is to minimize the discrepancy between the normal neighborhood features A_N^e and A_N^o and the reconstructed features $\hat{A}_N^e$ and $\hat{A}_N^o$. Therefore, we introduce the projection loss L_{pro}, which can be expressed as follows:

$$L_{\text{pro}} = \| \hat{A}_N^e - A_N^e \|_2^2 + \| \hat{A}_N^o - A_N^o \|_2^2 \tag{7}$$

where $\| \cdot \|_2^2$ is the mean square error, commonly used for training autoencoders. It is worth noting that autoencoders under the guidance of L_{pro} accomplish two proxy tasks: one is to inpaint the pseudo-anomalies in features, and the other is to predict the features of the neighborhood channels.

To supervise the pixel-level anomaly prediction of the AExNet, we introduce segmentation loss L_{seg} under the guidance of pseudo-abnormal mask, which can be represented as follows:

$$L_{\text{seg}} = L_{\text{Dice}}\left(M_{\text{pre}}, M_P\right) + L_{\text{CE}}\left(M_{\text{pre}}, M_P\right) \tag{8}$$

where $L_{\text{Dice}}(\cdot)$ denotes the dice loss [56] and $L_{\text{CE}}(\cdot)$ denotes the pixelwise binary cross-entropy loss. Since our proposed method follows an end-to-end training mode, the overall loss can be expressed as follows:

$$L_{\text{total}} = \lambda L_{\text{pro}} + (1-\lambda) L_{\text{seg}} \tag{9}$$

where $\lambda \in (0,1)$ is the loss weight.

IV. Experiments

In this section, our proposed method is validated on the MVTec AD [22] and Visa [31] datasets and compared with SOTA methods for image-level anomaly detection and pixel-level anomaly segmentation. To evaluate the effectiveness, ablation studies are conducted in terms of the network architecture and loss function. Furthermore, to demonstrate the

TABLE I

PIXEL-LEVEL ANOMALY SEGMENTATION RESULTS WITH AUC/AP (%) METRICS ON MVTEC AD DATASET. SUPERSCRIPT †: REPRODUCED RESULTS FROM AVAILABLE OFFICIAL IMPLEMENTATIONS. THE OPTIMAL AND SUBOPTIMAL RESULTS ARE SHOWN IN BOLD AND UNDERLINE, RESPECTIVELY. SUBSEQUENT TABLES FOLLOW THE CONSISTENT PRESENTATIONS

	Method → / Category ↓	DFR† NC'21	DRAEM† ICCV'21	DSR† ECCV'22	MemSeg† EAAI'23	SimpleNet† CVPR'23	DAF† TII'23	DiffAD ICCV'23	DeSTSeg† CVPR'23	MambaAD NIPS'24	SSMCTP TPAMI'24	GLAD ECCV'24	FC^{2}P (Ours)
Texture	Carpet	99.0/77.5	96.7/58.0	96.1/78.5	98.8/63.8	98.1/44.3	98.3/**79.3**	98.1/74.1	96.1/72.8	**99.2**/63.0	95.8/55.2	98.5/-	98.5/**80.1**
	Grid	98.3/30.4	99.7/69.4	99.6/67.1	97.3/45.1	98.6/33.0	99.3/56.5	99.7/73.7	99.2/48.9	**99.8/74.6**	**99.7**/69.7	99.6/-	99.7/73.2
	Leather	**99.7**/68.0	99.0/71.1	99.5/54.7	99.6/73.6	99.2/41.7	94.7/62.4	99.1/73.7	**99.8**/74.6	99.3/47.5	97.6/65.5	**99.8**/-	99.7/**76.9**
	Tile	99.3/93.6	99.2/95.1	98.5/94.6	98.9/94.9	96.3/60.6	96.5/92.8	**99.4**/95.1	98.4/92.2	93.1/42.8	99.3/**95.7**	98.7/-	98.5/93.0
	Wood	97.1/77.4	96.8/79.7	91.5/68.4	94.3/78.7	93.8/45.2	96.7/78.9	96.7/80.0	97.7/80.3	93.9/44.2	94.8/75.6	98.4/-	**98.6/88.0**
Object	Bottle	98.9/84.9	99.1/88.7	98.8/90.0	98.6/89.3	98.0/69.6	98.5/87.4	98.8/87.4	99.1/89.7	98.8/79.7	99.2/89.9	98.9/-	**99.6/93.4**
	Cable	96.9/65.6	96.4/66.1	97.7/77.8	94.6/69.5	97.5/65.6	97.4/65.1	96.8/64.9	96.5/56.9	98.0/54.4	95.5/61.6	98.1/-	**98.6/79.8**
	Capsule	96.6/42.9	89.3/46.3	91.1/45.7	96.2/56.6	98.9/42.4	97.1/33.6	98.2/54.4	**99.2**/53.7	98.6/45.0	93.4/52.0	98.5/-	98.9/**61.3**
	Hazelnut	98.9/64.6	**99.7**/91.7	99.1/88.7	96.8/72.5	97.7/44.6	96.7/61.8	99.4/85.9	**99.7**/89.0	99.0/61.8	99.5/89.1	99.5/-	**99.7/92.8**
	Metalnut	**99.6**/97.0	99.5/96.5	94.1/70.5	98.6/94.5	8.6/89.6	99.2/93.8	99.1/94.4	99.0/93.9	97.1/78.3	99.3/94.7	98.8/-	**99.6/97.4**
	Pill	98.0/67.1	97.5/51.8	94.2/69.2	94.8/**81.7**	98.6/79.6	98.5/73.5	97.7/68.9	98.7/81.6	97.3/64.5	97.4/46.9	97.9/-	**98.8**/78.5
	Screw	98.9/40.2	99.5/**72.3**	98.0/53.8	95.6/51.4	99.3/35.7	98.7/38.9	99.0/58.5	98.7/59.4	99.5/52.2	99.5/70.1	99.1/-	**99.6**/58.9
	Toothbrush	99.1/64.9	98.4/52.6	99.5/76.3	99.3/67.2	98.5/40.9	92.7/57.5	99.2/70.1	99.4/**77.5**	99.0/51.1	99.0/69.0	99.4/-	**99.5**/76.4
	Transistor	89.8/51.7	84.6/44.9	80.2/35.8	94.1/**76.4**	**96.7**/67.5	89.3/52.9	93.7/60.2	90.3/69.1	96.8/69.7	89.1/45.8	96.2/-	92.4/68.6
	Zipper	**99.2**/72.2	97.7/72.3	98.4/72.8	98.0/78.6	98.8/65.3	98.6/62.7	99.0/77.8	98.1/55.6	**99.2/85.1**	99.0/76.5	97.9/-	**99.2**/78.1
	Average	98.0/66.5	96.9/70.4	95.8/69.6	97.0/72.9	97.7/55.0	96.8/66.5	98.3/74.6	98.0/75.7	97.8/57.2	97.2/70.5	98.6/63.8	**98.7/79.8**

practicality, visualization results are shown on the KSDD2 [57] and Magnetic Tile [58] datasets that are obtained from real-world scenarios. Finally, the efficacy of the proposed AExNet is demonstrated through the visualization of abnormal features at different stages.

A. Experimental Setups

1) Datasets: The proposed method is validated on two popular and challenging anomaly detection datasets. Furthermore, to demonstrate its effectiveness and practicality, we perform visual validation on two datasets derived from real-world industrial scenarios. All four datasets provide precise pixel-level ground truth for validating anomaly segmentation performance.

1) *MVTec AD* [22] comprises 5354 images distributed over ten categories of objects and five categories of textures. Each category consists of approximately 200 normal images for training and 100 defective images for testing. The resolution of original images ranges between 700 × 700 and 1024 × 1024 pixels.
2) *Visa* [31] contains 10 821 images, including 9621 normal samples and 1200 abnormal samples, which are divided into 12 distinct subsets. Four of the subsets belong to the PCB category, and their structures are relatively complex. In some categories, such as capsules, multiple objects together constitute the target to be detected. Hence, this dataset is more challenging for anomaly detection and segmentation.
3) *Magnetic Tile* [58] originates from a real industrial scene, the largest magnetic tile production base in Zhejiang, China. It contains 1344 images, including 952 normal samples. The defect samples are divided into five subcategories, including blowhole, break, crack, fray, and uneven, which are defects on the magnetic tiles. The images exhibit large resolution differences and have different aspect ratios. In addition, the brightness of the images, the shape of the defects, and the texture of the tiles are also inconsistent.
4) *KSDD2* [57] is a surface defect detection dataset with over 3000 images, obtained while tackling a real-world industrial problem. The image resolution is approximately 230 × 630 pixels. It contains 2979 normal images and 356 defective images with several different types of defects (scratches, minor spots, and surface imperfections)

2) Evaluation Protocols: As in previous studies [6], [7], [59], the area under the receiver operating characteristic curve (AUC), the AP, and area under the per region overlap (AUPRO) metrics are considered for evaluating the pixel-level anomaly segmentation task. In addition, the AUC metric is also employed to detect image-level anomalies. Both evaluation protocols distinguish normal from abnormal instances by thresholding anomaly scores, with the AP focusing more on the accuracy of segmentation within abnormal regions.

3) Implementation Details: The resolution of images and masks is resized to 256 × 256 for both the training and testing phase. The Swin Transformer [53] pretrained using the ImageNet dataset serves as the feature extractor. The loss weight λ is set to 0.5 by default. Adam optimizer is applied with a learning rate of 1e^{-4}. The image-level anomaly scores are obtained following the approach in [6]. Our models are trained in an end-to-end mode for 400 and 200 epochs on MVTec AD and Visa, respectively, using a batch size of 8 on a single NVIDIA GTX 3090Ti, following the default PyTorch 2.2.1 framework.

B. Comparisons With SOTA Methods

This section presents quantitative results of the proposed method on the MVTec AD and Visa dataset in detail, verifying the superiority of our method over SOTA methods.

1) Results on MVTec AD Dataset: Table I shows the pixel-level AUC and AP results for anomaly segmentation on the MVTec dataset, while Table II showsthe pixel-level AUPRO results. The baseline methods compared include DFR [7], DRAEM [6], DSR [48], MemSeg [30], SimpleNet [59], DAF [51], DiffAD [49], DeSTSeg [12], SSMCTP [60], MambaAD [61], and GLAD [62]. The results show that the

TABLE II
COMPARISON OF PIXEL-LEVEL ANOMALY SEGMENTATION WITH AUPRO (%) ON MVTEC AD AND VISA DATASETS

Method	DFR†	DRAEM†	DSR†	MemSeg†	SimpleNet†	DAF†	DiffAD	DeSTSeg†	MambaAD†	SSMCTP	GLAD	Ours
MVTec AD	87.5	92.8	90.8	93.2	89.6	92.7	-	94.7	94.0	-	-	**96.5**
Visa	64.3	73.1	68.1	75.6	68.9	81.2	-	89.9	91.9	-	-	**92.8**

TABLE III
COMPARISON OF IMAGE-LEVEL ANOMALY DETECTION WITH AUC (%) ON THE MVTEC AD DATASET. RESULTS ARE AVERAGED OVER ALL CATEGORIES

Method	DFR†	DRAEM†	DSR†	MemSeg†	SimpleNet†	DAF†	DiffAD	DeSTSeg†	MambaAD	SSMCTP	GLAD	Ours
AUC	93.5	98.0	98.2	**99.6**	**99.6**	97.6	98.7	98.4	98.7	97.2	99.3	98.9

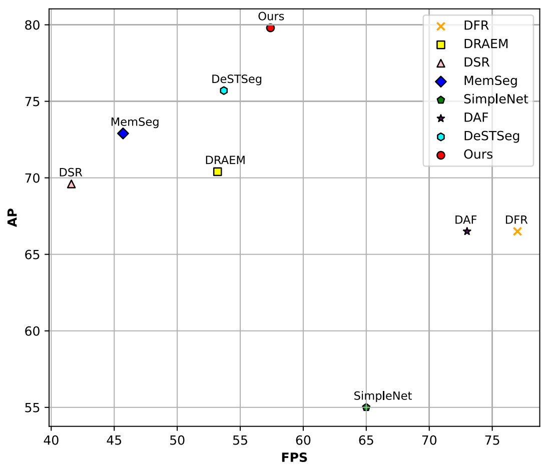

Fig. 5. Comparison of inference speed (FPS) versus pixel AP for different methods on MVTec AD benchmark.

TABLE IV
RESULTS OF ANOMALY DETECTION AND SEGMENTATION WITH AUC AND AP (%) METRICS ON THE VISA DATASET, COMPARING WITH SEVERAL SOTA METHODS. RESULTS ARE AVERAGED OVER ALL CATEGORIES

Method	Det.	Seg.	
	AUC	AUC	AP
DFR†	92.6	97.8	34.7
PatchCore†	88.9	96.1	36.7
DRAEM†	93.9	93.2	33.5
SimpleNet†	95.0	**98.1**	36.0
DAF†	91.8	96.2	35.6
D3AD	94.1	97.9	-
DeSTSeg†	90.6	93.8	36.5
MambaAD	**95.3**	97.8	40.4
Ours	92.9	98.0	**44.8**

proposed method outperforms all current SOTA methods for the anomaly segmentation task and achieves a new SOTA on the MVTec AD dataset, obtaining 98.7% AUC and 79.8% AP. Focusing on AP protocols that better evaluate segmentation performance, our method improves the score by a significant 4.1↑ compared with the current SOTA knowledge distillation-based method DeSTSeg with two-stage training mode and by 13.0↑ compared with the SOTA image reconstruction-based method GLAD using a high computational cost latent diffusion model. It is noteworthy that the proposed method achieves optimal results in nine categories for AUC and eight categories for AP, which is comparable to the SOTA method in other categories, as well as yielding the best AURPO result. Furthermore, Table III shows the image-level AUC results of our method for anomaly detection on this dataset. In addition, the frames per second (FPS) for inference speed versus pixel AP is shown in Fig. 5. Our method strikes a tradeoff between speed and performance, highlighting its adaptability for practical applications. The results demonstrate that the proposed method achieves comparable performance to current SOTA methods. In summary, the evaluation of anomaly segmentation and anomaly detection validates the effectiveness and robustness of our proposed method.

2) Results on Visa Dataset: Table IV reports the AUC and AP average results over all categories for image-level anomaly detection (Det.) and pixel-level segmentation (Seg.), and Table II shows the pixel-level AUPRO results on the Visa dataset. The baseline methods compared include DFR [7], PatchCore [8], DRAEM [6], SimpleNet [59], DAF [51], D3AD [63], DeSTSeg [12], and MambaAD [61]. For the anomaly detection task, our proposed method achieves performance comparable to SOTA methods. Considering the anomaly segmentation results, our method is on par with the SOTA method SimpleNet under the AUC. However, for the AP, our method shows clear advantages, improving upon SimpleNet by 8.8↑. SimpleNet adds Gaussian noise to the features to simulate anomalies and uses a simple binary classifier for detection. In contrast, our method generates more precise pseudo-anomalies in the feature and utilizes the proposed AExNet to mine anomaly representations in the multistage features, resulting in more accurate anomaly score maps. For AUPRO, our method outperforms the current SOTA methods DeSTSeg and MambaAD by 2.9% and 2.5%, respectively. Notably, the proposed method outperforms most previous methods on the precise anomaly segmentation task. Previous SOTA methods averaged around 36.0% AP, while our method reaches 44.8%, outperforming the SOTA method PatchCore by 8.1%↑. These experimental results strongly demonstrate the excellent performance of our method in anomaly segmentation and also demonstrate its effectiveness and generalization capabilities.

3) Visualization Results: We present comprehensive qualitative results on the MVTec AD dataset to compare the anomaly segmentation performance of our method with several

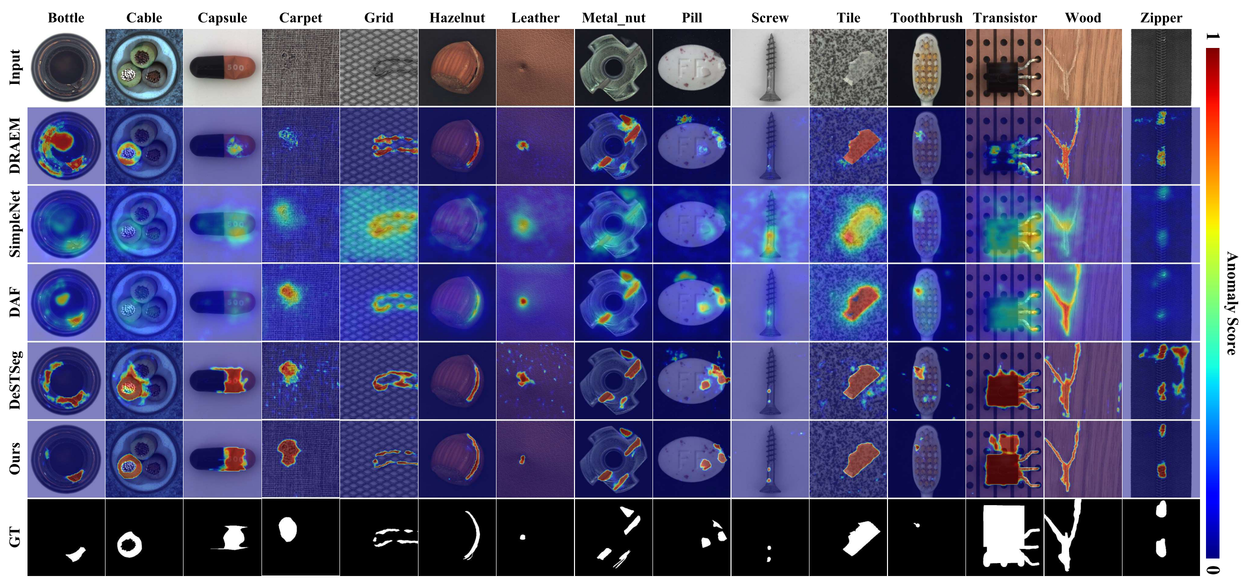

Fig. 6. Qualitative visualized results for anomaly segmentation. The last row is the ground truth of the given input abnormal image. Compared with the baseline methods, including DRAEM (second row), SimpleNet (third row), DAF (fourth row), and SOTA DeSTSeg (fifth row) on MVTec AD datasets, our proposed approach FC^{2}P (sixth row) has a more accurate and compact anomaly location capability. Specifically, the superiority of our segmentation results is manifested in the suppression of false positives within normal backgrounds and the accurate delineation of the contours of actual anomalies.

SOTA methods, as shown in Fig. 6. Our method outperforms SOTA methods, yielding more accurate and significant anomaly segmentation results that closely align with the ground truth. Specifically, our method excels in two key aspects: 1) accurately delineating abnormal regions with well-defined boundaries and 2) reducing false positives in normal regions. For example, our method accurately segments the color anomalies in the cable's circular periphery without detecting the inner core. Notably, the anomaly segmentation maps generated by our method for the carpet are more accurate than the provided annotations, closely matching the actual anomaly shapes.

Visualizations of the Visa dataset further demonstrate the superiority of our method in anomaly segmentation. The defects in the dataset are often small and challenging to detect, as shown in Fig. 7. Furthermore, some categories exhibit complex textures, such as the PCB. Despite these challenges, our method achieves superior anomaly segmentation, with more comprehensive and meaningful segmentation results and fewer false positives in both object and background areas.

Fig. 8 shows visualizations of several cases from the real magnetic tile defect and KSDD2 datasets. The predicted anomaly score map closely matches the true label. Notably, the true label in the second column does not mark the small defect at the bottom (likely missed), whereas our proposed method detects it. The results demonstrate the effectiveness and practical implications of the proposed method.

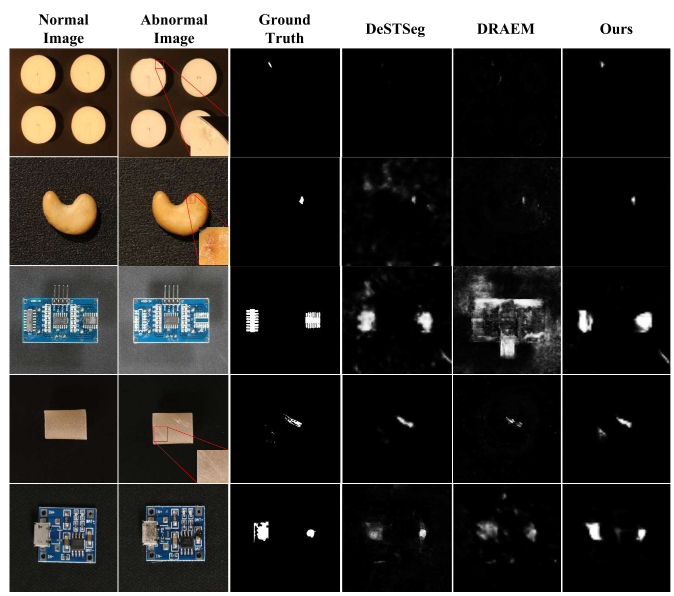

Fig. 7. Qualitative results of anomaly segmentation on the challenging Visa dataset, comparing with DRAEM and SOTA method DeSTSeg. The categories from top to bottom are, respectively: candle, cashew, pcb1, pipe_fryum, and pcb4. The proposed method demonstrates significant segmentation capabilities on objects with complex structures as well as on small defects, and it generates fewer false positives in backgrounds.

C. Ablation Study

This section analyzes the contributions of the components of the proposed method.

1) Influence of Different Components: Table V presents the ablation studies on the MVTec AD dataset to assess the effectiveness of each component of the proposed method. DFR [7] is chosen as our baseline, and various capabilities are gradually added, resulting in the following six experiments: 1) adding inpainting ability (inp.); 2) adding prediction ability (pre.); 3) adding inpainting and prediction ability as complete FC^{2}P; 4) adding SAP module without F_{all}; 5) using vanilla encoder in U-Net to segment F_{all}; and 6) training with all proposed components. The baseline method exhibits substantial improvement when the capabilities of two nonidentity reconstructions are added (compared with 66.5 in Table I). Furthermore, we perform a density analysis of the pixel anomaly scores between the baseline and our proposed FC^{2}P, as shown in Fig. 9. It can be observed that our proposed

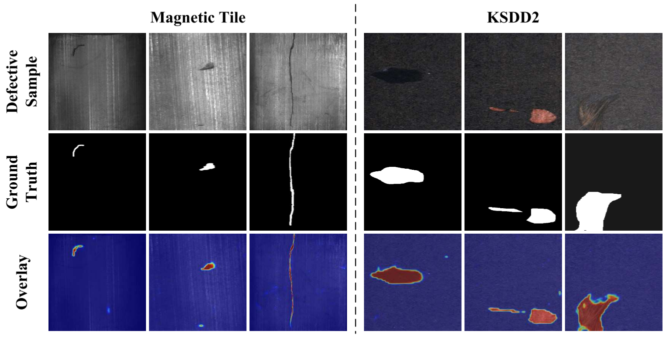

Fig. 8. Visualization results on the Magnetic Tile (left) and KSDD2 (right) datasets.

TABLE V

ABLATION STUDIES FOR ANOMALY DETECTION AND SEGMENTATION WITH DIFFERENT COMPONENTS OF OUR PROPOSED ARCHITECTURE ON MVTEC AD DATASET. RESULTS ARE REPRESENTED AS THE AVERAGE AP SCORE OVER ALL CATEGORIES

#	FC^2P		AExNet		Pixel AP
	inp.	pre.	SAP	F_{all}	
1	✓				67.5
2		✓			68.7
3	✓	✓			69.3
4	✓	✓	✓		78.5
5	✓	✓		✓	<u>79.6</u>
6	✓	✓	✓	✓	**79.8**

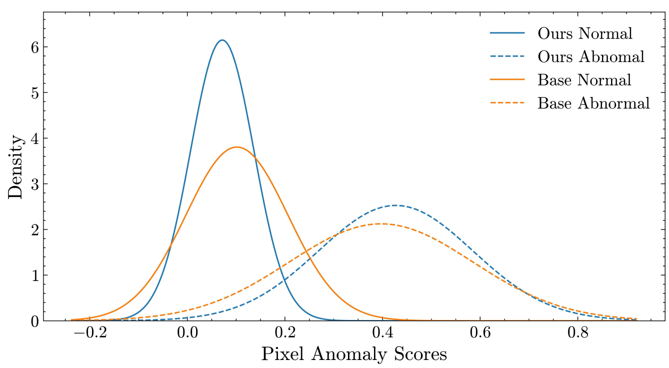

Fig. 9. Probability density of both normal and abnormal pixel anomaly score for feature reconstruction residual with FC^{2}P, comparing with the base method.

method achieves a more compact distribution with less overlap for both abnormal and normal pixels, making it easier to distinguish them. These results show that traditional feature reconstruction-based methods suffer from overgeneralization. When the AExNet is added, anomaly segmentation performance improves significantly by at least 9% AP. Moreover, our approach improves by +10.3%↑ when using F_{all} with vanilla encoder, indicating that it actually contains rich abnormal information. When the SAP module is added with F_{all}, our proposed method achieves the optimal result. Based on the divide-and-conquer approach, the gradual enhancement of anomalies at different stages, combined with the integration of overall feature reconstruction residuals, significantly improves anomaly segmentation performance.

2) Influence of Loss Function: Fig. 10 illustrates the effect of loss weights on the performance of anomaly detection and segmentation on the MVTec AD dataset. The loss weight

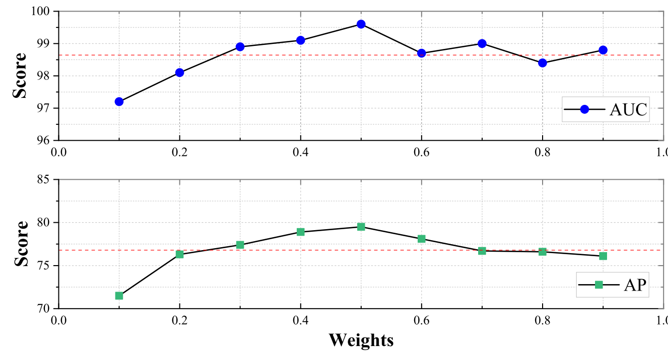

Fig. 10. Performance with varied loss weights on MVTec AD for anomaly detection (AUC) and segmentation (AP). The red dashed line indicates the mean value. Optimal performance is achieved when the reconstruction loss and segmentation loss are balanced. Furthermore, adjusting the weights can be employed to favor different tasks.

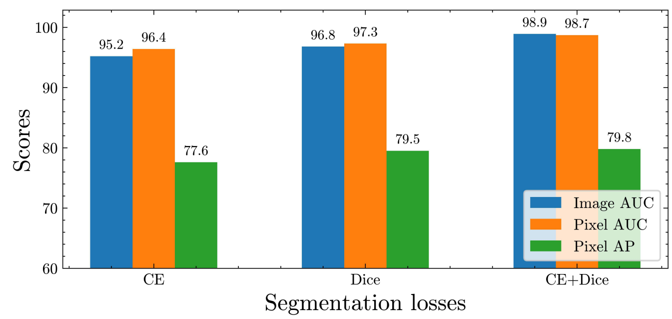

Fig. 11. Performance with segmentation loss components for anomaly detection and segmentation in all metrics. The combination of both CE and dice losses indeed compensates for their respective deficiencies. Not only maintains the accuracy of anomaly pixel classification in unbalanced classes but also enhances the model's sensitivity to abnormal regions.

λ controls the balance between reconstruction loss and segmentation loss during end-to-end training, thereby affecting the resulting anomaly score maps. As the reconstruction loss weight gradually increases to 0.4, the performance of image-level anomaly detection improves from 97.1% to 99.1%. Beyond this threshold, the feature reconstruction module FC^{2}P tends to overfit, resulting in a plateau in anomaly detection performance. However, as the proportion of segmentation loss decreases, there is a noticeable decline in the AP metric for anomaly segmentation. This suggests that the AExNet, designed to mine anomalies from feature residuals, is inadequately trained. Although the feature reconstruction component is well-trained, the segmentation network fails to extract anomalies from the features to produce accurate segmentation results. The results indicate that optimal performance is achieved when the reconstruction loss and segmentation loss are properly balanced. In addition, it is observed that reconstruction loss is more conducive to image-level anomaly detection, making it beneficial for adapting to various task types.

Furthermore, ablation analysis is conducted on the components of the segmentation loss, as shown in Fig. 11. Since the CE loss focuses on independent pixel predictions, it performs poorly when used alone in class-imbalanced segmentation tasks, particularly in anomaly segmentation where normal pixels vastly outnumber abnormal pixels. In contrast, the dice

TABLE VI

ABLATION STUDIES FOR ANOMALY DETECTION AND SEGMENTATION WITH DIFFERENT BACKBONES ON MVTEC AD AND VISA DATASETS

Backbone	MVTec			Visa		
	Image AUC	Pixel AUC	Pixel AP	Image AUC	Pixel AUC	Pixel AP
ResNet-18	98.5	98.1	76.0	92.4	96.2	38.0
ResNet-50	97.3	97.6	78.5	91.5	96.8	39.6
WRN-50	97.6	97.9	77.4	91.0	97.1	40.2
DenseNet-121	97.8	98.4	77.1	91.5	96.4	41.8
PVT-v1	98.0	98.0	78.3	91.9	97.6	40.5
HorNet	98.3	98.1	78.6	92.1	97.8	42.1
Swin-base	98.1	98.6	77.6	91.5	97.8	43.1
Swin-large	**98.9**	**98.7**	**79.8**	**92.9**	**98.0**	**44.8**

TABLE VII

ABLATION STUDIES FOR ANOMALY DETECTION AND SEGMENTATION WITH DIFFERENT COMBINATIONS ON MVTEC AD DATASET

Stage 1	Stage 2	Stage 3	Image AUC	Pixel AUC	Pixel AP
✓			95.2	96.8	75.6
	✓		96.3	97.5	78.3
		✓	97.6	95.6	73.8
✓	✓		98.5	98.6	79.1
	✓	✓	98.6	98.1	78.8
✓	✓	✓	**98.9**	**98.7**	**79.8**

loss focuses on the overall structural similarity of the abnormal regions, which alleviates the class-imbalance issue to some extent; however, the lack of confidence in pixel classification results in an AP of only 79.8%. Interestingly, combining these two types of losses compensates for their respective limitations, not only maintaining the accuracy of abnormal pixel classification but also increasing the model's sensitivity to abnormal regions, thereby significantly improving anomaly segmentation performance.

3) Influence of Pretrained Network: Table VI illustrates the impact of different pretrained backbones, including CNN-based (ResNet [38], WRN [64], and DenseNet [65]) and Transformer-based (PVT [66], HorNet [67], and Swin [53]) models with their versions, on the performance of anomaly detection and segmentation. The results on the MVTec AD and Visa datasets demonstrate the significant advantage of backbones following the Swin Transformer framework in both anomaly detection and segmentation tasks. First, the Swin Transformer not only captures long-range dependencies, a characteristic absent in CNNs, but its hierarchical structure also facilitates the processing of multiscale information. This is essential for understanding both the macrostructure of normal images and their fine-grained details. In addition, the window-based self-attention mechanism enhances the model's ability to bridge local and global contexts, facilitating the accurate identification and localization of abnormal regions.

4) Influence of Feature Stages: Table VII illustrates the impact of combining different feature stages on anomaly detection and segmentation performance in the MVTec AD dataset. The results demonstrate that optimal performance can be achieved by combining features from all three stages. Notably, the combination of stages 1 and 2 is more beneficial for segmentation, while the combination of stages 2 and 3 is more beneficial for detection. The features from the second

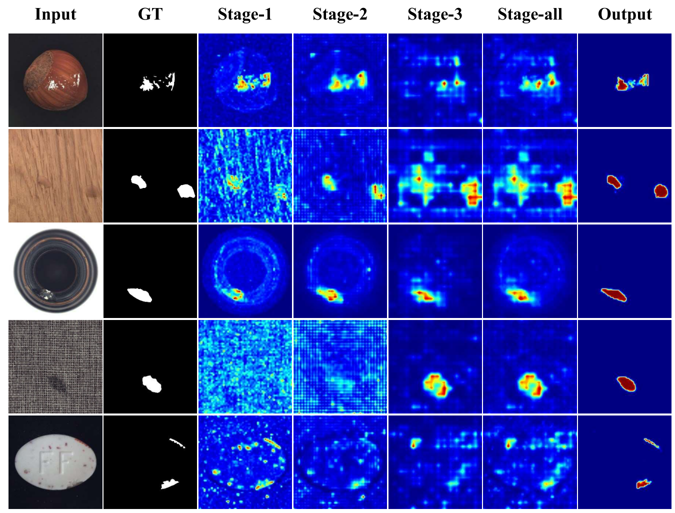

Fig. 12. Some examples of visualization of different feature residual stages on MVTec AD dataset. The sixth column represents the results of aggregating the three stages. The proposed method can exploit the advantages of all stages and produce precise anomaly score maps, eliminating false positives in normal areas.

stage play a key role, achieving relatively good results with just this stage. This is because shallower features tend to focus on detailed information in abnormal regions, while deeper features are more concerned with semantic aspects. Interestingly, the SOTA methods SimpleNet and Patchcore both utilize stages 2 and 3 to achieve optimal performance. However, the anomaly score map is often inaccurate due to the absence of shallow information. It can be observed that our proposed method fully exploits the characteristics of all feature stages, achieving optimal results in both detection and segmentation. This is primarily due to the fact that the AExNet not only considers anomalies as a whole but also enables anomalies to gradually enhance across the stages of feature reconstruction residuals.

Furthermore, to validate that our method addresses the issue of anomalies being overshadowed at different feature stages, we visualize the feature reconstruction residuals and their aggregation for anomaly segmentation in Fig. 12. We observe that anomaly segmentation performance varies across different stages and categories. For instance, the print anomaly on the hazelnut is more prominent in stage 1, but aggregating all levels can obscure this detail. Similarly, the two anomalies on the pill are suppressed by false positives in stage 3, rendering them insignificant in the aggregated results. Our method alleviates this issue by reducing false alarms in normal areas and producing clear outlines of abnormal areas. Notably, even when the shallower stages (stages 1 and 2) fail to detect the color anomaly on the carpet, our method can still extract weak details to assist deeper stages in generating a precise anomaly score map.

V. CONCLUSION

This article proposes a novel feature reconstruction-based framework named FC^{2}P to address the challenging and practical task of unsupervised anomaly segmentation. The proposed method primarily alleviates the issues of overgeneralization and overwhelming that are typically encountered in traditional feature reconstruction-based approaches. Specifically, features are split into two subsets based on neighboring channels, and two autoencoders are employed for their closed-loop prediction, driven by our observation of the potential relationship between neighboring channel feature maps in the pretrained features. In addition, for the feature reconstruction residual subsets, an AExNet is proposed, adopting the divide-and-conquer principle, so that anomalies are gradually enhanced across feature stages and combined with the aggregated feature residuals to decode a precise anomaly score map. The proposed method outperforms SOTA methods on two widely used benchmark datasets in both anomaly detection and anomaly segmentation. Finally, the superiority and practicality of the proposed method in anomaly segmentation are thoroughly demonstrated through visual qualitative analysis.

Limitations: Although the proposed method achieves strong performance, it sacrifices some inference efficiency. This is primarily caused by the aggregation of multistage pretrained features, which increases the number of channels and leads to significant computational overhead during the reconstruction process of the autoencoders.

Future Work: In future work, we plan to integrate knowledge from large vision-language models [68] or vision foundational models [69] into our approach. This integration aims to enhance the generalization capability of our method for anomaly detection and segmentation tasks.

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